

Assignment: Frequent Itemsets, Apriori & FP-Growth with Association Rules

Transaction Dataset (12 Transactions)

T1: Milk, Bread, Butter

T2: Bread, Diapers, Beer, Eggs

T3: Milk, Bread, Diapers, Beer

T4: Bread, Milk, Diapers, Beer

T5: Milk, Bread, Butter

T6: Bread, Butter

T7: Milk, Diapers, Beer

T8: Milk, Bread, Diapers, Butter

T9: Bread, Diapers, Beer

T10: Milk, Bread, Beer

T11: Milk, Bread, Diapers

T12: Bread, Butter, Eggs

Part 1: Apriori Algorithm

Minimum Support = 3

1. Generate C1 and compute support.
2. Generate L1.
3. Generate C2 and compute support.
4. Generate L2.
5. Generate C3 and compute support.
6. Generate L3.
7. Stop when no more frequent itemsets.
8. Provide all frequent itemsets with support.

Part 2: Association Rules (Confidence & Lift)

Minimum Confidence = 60%

1. Generate association rules from frequent itemsets.
2. Compute confidence for each rule.
3. Prune rules below threshold.
4. Compute lift for strong rules.
5. Interpret lift (>1 positive, $=1$ independent, <1 negative).
6. Provide rule, support, confidence, lift, interpretation.

Part 2: FP-Growth Algorithm

Minimum Support = 3

1. Build frequency table.
2. Construct FP-tree.
3. Provide all frequent itemsets.

Part 4: Comparison & Reflection

1. Compare Apriori and FP-Growth.
2. Which performs more scans?
3. Identify most frequent item and strongest pair.
4. Why does FP-Growth avoid candidate explosion?